

HOLC Redlining Maps and Present-Day Geography

A form filled in on the edge of Cleveland in 1939, and the ground it still describes

84-355 Democracy's Data

August 2026

An appraiser working for the federal government walked a few streets on the southern edge of Cleveland in the late summer of 1939 and then filled in a form. The form had spaces for the terrain, the class of the inhabitants, the age and condition of the buildings, the availability of mortgage money. Under the heading *Clarifying Remarks* he wrote:

This area is the least desirable neighborhood in Garfield Heights. Immediate proximity to negro settlement abutting north boundary – absence of transportation – lack of sewers – unpaved streets and general isolation all have taken their toll. This community was developed about 20-25 years ago and then stopped with the influx of negroes into abutting area. At present, property can be sold only at sacrifice and mortgage money is not available which indicates that potentially this neighborhood will eventually be taken over by negroes with improvement in their financial condition. Approximately 30 homes are located in this area; developed about 40%; no further development is anticipated for white ownership.

The area is D9, *Garfield Hts. (Orchard Ave. Section)*, and the sheet is dated 9-9-39. Two other lines on the same form matter as much as the paragraph. One is headed *Infiltration of*, and on it he wrote *Potential influx of Negro*. The other asks for the proportion of Black residents already living in the area, and on that line he wrote 0 – none.

Read those two lines together and the reasoning is not hidden, inferred, or reconstructed by a later generation. **Race was not a proxy for something else in these documents. It was the stated criterion**, applied here to a neighborhood where, by the appraiser's own count, no Black family yet lived. The last sentence he wrote is the one to carry forward: *no further development is anticipated for white ownership*.

He gave the area a grade of D.

01 Best, Still Desirable, Definitely Declining, Hazardous

That grade was not an opinion filed in a drawer. Congress had created the **Home Owners' Loan Corporation** in 1933 to refinance mortgages during the Depression. To decide where lending was safe, the HOLC produced "residential security maps" for more than two hundred American cities. Every neighborhood in a mapped city was colored one of four grades:

Grade	Label	Color
A	Best	green
B	Still Desirable	blue
C	Definitely Declining	yellow
D	Hazardous	red

The D areas were printed in red. That is where the word *redlining* comes from, and the paragraph above is what a D looked like from the inside.

Cleveland was surveyed in 1939. Its sheet covers 190 graded areas over 302 square kilometers, and the whole of it is drawn in Figure 1 from the digitized polygons, filled with the four colors the corporation itself used. The color of each area is the one recorded in the source file; nothing here is a palette chosen for effect.

HOLC grade, 1939

- A Best (31 areas, 11% of graded land)
- B Still Desirable (55 areas, 23% of graded land)
- C Definitely Declining (76 areas, 47% of graded land)
- D Hazardous (28 areas, 18% of graded land)

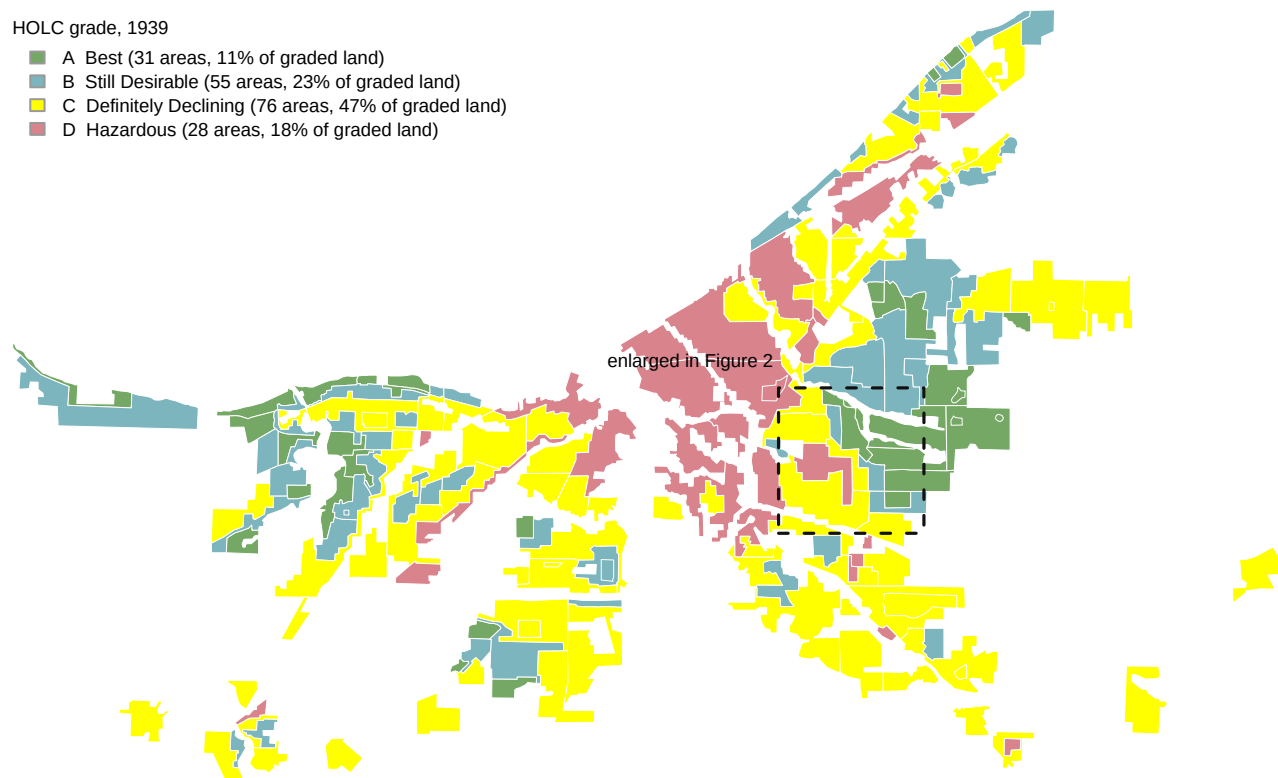


Figure 1. Cleveland as the Home Owners' Loan Corporation graded it in 1939. 190 graded areas over 302 square kilometers, filled with the corporation's own four colors. Grade D, "Hazardous", is red and sits an average of 7.8 km from the center of the surveyed area; grade A, "Best", is green and sits 12.4 km out. Grade C alone covers 47% of the graded land. The dashed rectangle marks the ground enlarged in Figure 2.

The shape of the thing is a finding in itself, before any census data arrives. Red is in the industrial core and along the river; green is out at the edge; a broad band of yellow lies between them. Nobody drawing this map was recording an accident of geography. They were drawing concentric rings of creditworthiness around a downtown, and the ring you lived in decided whether a bank would lend on your house.

02 How a 1939 opinion becomes a file you can open

Nothing in Figure 1 is a reconstruction. The corporation drew those lines, its appraisers filled in those forms, and the sheets survived as federal records because a federal agency was obliged to keep them. What turned them into something a chapter can compute with is a second act of labour, seventy years later.

The Digital Scholarship Lab at the University of Richmond photographed the surviving sheets, traced every polygon off them, transcribed the area description forms, and published the result as **Mapping Inequality** under a Creative Commons license. The national file was fetched here on 2026-08-10: 10.6 MB of GeoJSON covering 314 city surveys. The paragraph at the top of this chapter is not a paraphrase either. It is what the appraiser wrote on sheet D9 of the OHCleveland1939 survey, printed from the same archive without a word changed.

That lineage is the reason the evidence is worth anything. This is not a survey of what people believed about neighborhoods in 1939, with all the sampling trouble that would carry. It is the working record of the agency doing the sorting. It was written to be acted on rather than read later, by people with no motive to overstate the role of race, because they did not think it needed justifying. A researcher's after-the-fact index of "which neighborhoods were disfavored" would be a weaker instrument, and would smuggle in the researcher's own theory of what disfavor looks like.

The defects run in two directions. Some belong to the file. Of the 10,154 digitized areas, 830 carry no grade at all. And a hand-drawn 1930s boundary, traced from a photograph of an aging sheet, is precise to a block rather than to a parcel.

Others belong to the quantity itself, and no better file would fix them. The HOLC surveyed cities above a population threshold, so there is no national frame here and never was one. A grade is one appraiser's judgment recorded as a letter; it has no standard error and no second rater. And the thing this chapter wants to measure, what a 1939 line did to the ground, is not a quantity anyone recorded. The people who would have to be counted moved, died, and were replaced by others over eight decades.

The file also needed handling before it could be trusted. The grade lives in a property called `grade`; the more natural guess, `holc_grade`, exists in other distributions of these maps and returns nothing here, which would have produced a complete, silent, entirely ungraded map. The per-city archives are addressed by a key built from state, city and survey year, and a wrong year does not return an error. The server answers with a page shell and a success code, so each response is checked by parsing it rather than by reading its status. On the census side, 4,489 tract records across 6 states carry an eleven-digit GEOID that has to be read as text; read as a number, every California identifier loses its leading zero and matches nothing. None of that is visible in a finished table, which is the point of saying it out loud.

03 A form with no field names, and a polygon measured in degrees

The paragraph this chapter opens with came out of a file. Here is the file. The national download is GeoJSON, one feature per HOLC area, with geometry and properties together. These are the properties of the Cleveland area the appraiser was writing about, area D9:

Property	What it holds	Value
area_id	identifier for this HOLC area	378
city_id	identifier for the city it sits in	115
grade	the HOLC grade, A to D — and note it is not called holc_grade	"D"
fill	the colour the HOLC itself printed the polygon in	"#d9838d"
label	the area's label on the original sheet	"D9"
name	a name field, blank on this record	" "
category_id	the grade again, as a number: the same fact in a second vocabulary	4
sheets	how many appraisal sheets survive for this area	1
area	the polygon's area in square degrees, which is not a unit of anything	2.81294353184514e-05
bounds	the bounding box, as two latitude-longitude corners	[[41.42575, -81.58124], [41.43494, -81.57687]]
residential	whether the appraiser marked the area residential	true
commercial	whether commercial	false
industrial	whether industrial	false
geometry	the boundary itself, as nested coordinate lists	MultiPolygon, coordinate lists nested four deep

Three things in those fourteen fields are worth stopping on.

The grade is called grade. Other distributions of these same maps call it `holc_grade`, and asking this file for `holc_grade` does not fail. It returns `NULL` for every one of the 10,154 features, which would produce a complete, plausible, entirely ungraded map. A misspelled column name is the quietest bug in this book.

area is 2.81e-05. That is not an error, and it is not acres. It is the shape's area in square *degrees*, which is not a unit of anything, because a degree of longitude is a different distance in Cleveland than in Birmingham. Every area, share and overlap in this chapter is computed after reprojecting to metres. A reader who summed that column would get a number, and the number would mean nothing.

category_id is 4 and grade is D. The same fact twice, in two vocabularies, in one record — and a third, `fill`, holds the color the HOLC itself printed the polygon in. This chapter draws its maps in that color rather than picking one, so the palette is evidence rather than decoration.

Now the other file, and it is the strangest object in this chapter. The appraiser's area description survives as JSON, and the JSON has no field names:

Key	Value
0	Greater Cleveland
1.a.1	(empty)
1.a.2	Very slowly
1.a.3	(empty)

Key	Value
1.b	Laborers - factory workers
1.c.1	65
1.c.2	Slovaks - few Germans
1.d	0
1.e	Potential influx of Negro
2.a.1	4-6 rm. singles
2.a.2	(empty)
2.b.1	Frame
2.b.2	(empty)
2.c.1	20 yrs.
2.c.2	(empty)
2.d.1	Fair
2.d.2	(empty)
...	...

Those keys are positions on a paper form. "1" is the section headed *Inhabitants*; "e" within it is the line the form calls *Infiltration of*, and what the appraiser wrote on that line is `Potential influx of Negro`. The transcription is faithful precisely because it kept the form's own structure instead of interpreting it, which means **the file cannot be read at all without the blank form beside it**. The digitizers made the right call and it is why the quotation at the top of this chapter can be printed without a word changed.

Against those two, the clean file is almost aggressively plain:

GEOID	State	City	Grade	Total	White	Black	Pct black
39035101101	OH	Cleveland	C	1686	670	705	41.81
39035101102	OH	Cleveland	B	4338	3282	468	10.79
39035101300	OH	Cleveland	C	1638	937	432	26.37

One row per census tract: a text identifier, a state, a city, a letter, three counts and a percentage. Every trace of the objects above is gone. **The polygon, the sheet, the appraiser's sentences and the eighty-one years in between have all collapsed into one letter in one column.** That collapse is what makes the rest of this chapter possible, and what its final section is careful about.

Two of those columns deserve their names read closely. GEOID is text, not a number, and this chapter's setup says so explicitly: 4,489 tract records across 6 states, of which the Californian ones begin with a zero that a numeric read silently deletes. And `grade` is not a measurement. It has no standard error, no second rater and no units; it is one man's letter, and it is being used here as though it were a treatment assignment. The chapter's closing section is about exactly how far that can be pushed.

04 A map drawn to allocate credit ends up allocating civic life

Homeownership is how most American families hold wealth, and mortgage access is how they get it. A policy that made credit unavailable in specific neighborhoods for decades did not only affect who lived where. **It affected who could accumulate anything to pass on.**

From there the consequences run through nearly every part of governing that happens on a map. School funding is property-tax based. Political participation tracks residential stability, and residential stability tracks whether people own. Municipal services, environmental burdens, transit investment and policing are all allocated geographically, using the same neighborhood boundaries that the credit market had already sorted. A document written to answer a narrow question about lending risk becomes, over decades, an input to almost everything a local government does – and this document was written by the federal government.

So the question the rest of this chapter asks is simple to state and unpleasantly hard to answer. Eighty years later, on the actual ground, does the map still show?

05 Two geographies that were never meant to meet

Answering it requires putting two things beside each other that have no reason to be compatible.

What	From	Unit	Vintage
HOLC grades	Mapping Inequality, Univ. of Richmond	Hand-drawn 1930s neighborhood polygons	1935–1940
Demographics	2020 Decennial Census	Census tracts	2020

These units do not correspond, and no amount of care will make them. A 1937 appraiser's neighborhood boundary was drawn with no knowledge of a 2020 census tract, and census tracts have themselves been redrawn every decade since. There is no shared identifier, no lookup table, and no reason the shapes should line up. A published tract-level redlining score does exist, but it sits behind a login, and inheriting a number from a file you cannot inspect is the worst of the available options.

So the link between the two geographies has to be built. The rule used here is the simplest one that can be defended: **take each census tract's center point and ask which HOLC polygon contains it.**

Step	Value
HOLC areas digitized	10,154
of them graded A to D	9,324
Rule applied	tract center point inside a graded polygon
Tracts matched to a grade	4,489
Cities represented	56

Step	Value
States	6

The rule fits in one sentence, and the sentence hides everything that can go wrong with it. Figure 2 shows it running on real ground: a 5.6 km window of Cleveland, with 1939 underneath in its own colors and 2020 drawn on top.

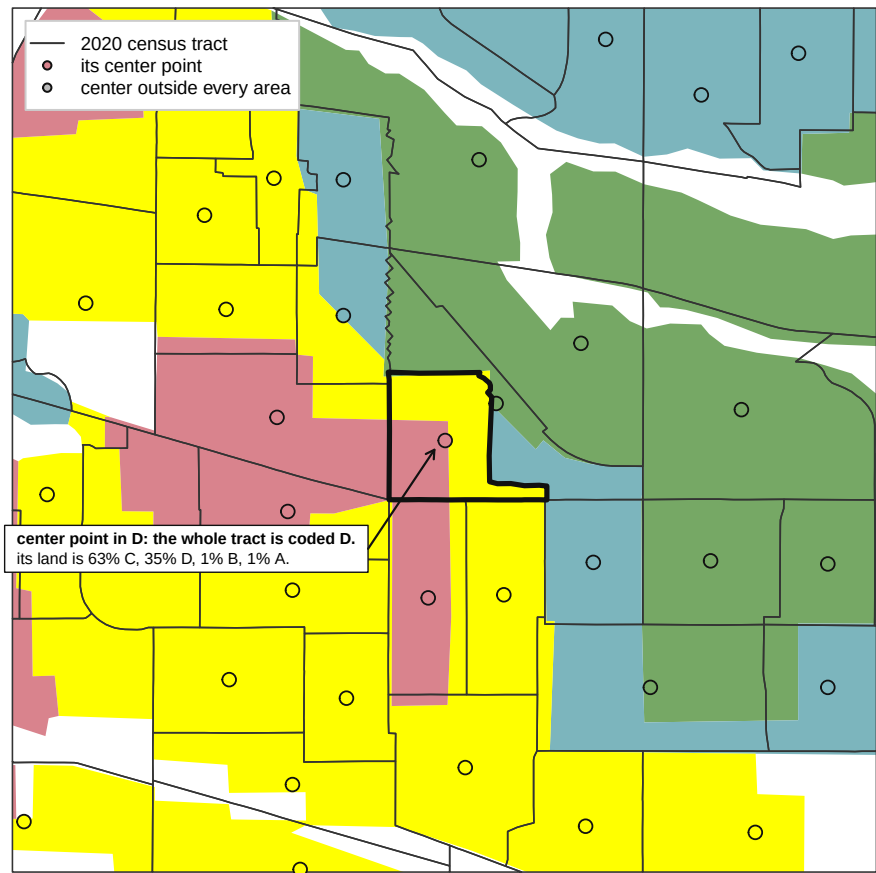


Figure 2. A 5.6 km window of Cleveland with 1939 underneath and 2020 on top. Colored surfaces are HOLC areas in the corporation’s own grade colors, and thin outlines are 2020 census tracts. Each dot is a tract center point, the only part of a tract the matching rule ever consults, colored by the grade that point returns. White gaps are land nobody graded.

The heavily outlined tract is the rule getting it wrong. Its center point falls in a D area, so the whole tract is coded D in every table in this chapter. But its land is 63% C, 35% D, 1% B, 1% A, and most of it was graded C.

Nothing lines up, because nothing was ever meant to. The colored surfaces were drawn by a federal appraiser in 1939. The outlines were drawn by the Census Bureau in 2020, for entirely unrelated reasons, from a different set of instructions, about a different question. Between them lies exactly one point per tract, and everything the rest of this chapter reports about eighty years of American housing rests on where that point happens to fall.

The outlined tract in Figure 2 was not chosen by eye. Among Cleveland tracts almost entirely covered by graded areas, involving a D area, and receiving a grade that is not the

grade covering most of them, it is the one where the disagreement is widest. And the rule has a second failure mode with no visible warning at all: **a tract whose center lands outside every graded area is dropped, however much of it overlaps one.** In Cleveland 103 tracts overlap a graded area and are dropped anyway, the largest of them 86% covered by 1939 grades.

06 The cost of the rule, measured

One window is an anecdote. Figure 3 puts the same question to every tract in the two cities this chapter goes on to name – two of the 56 in the file, not all of them. Of the graded land inside this tract, how much of it actually carries the grade the tract was given?

Before you read on, commit to a number. Figure 3 covers the 587 tracts in Cleveland and Philadelphia that the center-point rule managed to grade at all. **How many of those 587 were handed a grade that is not the grade covering most of their own land** – the failure the outlined tract in Figure 2 is one instance of? Write down a count, or a share of 587, before you look at the figure.

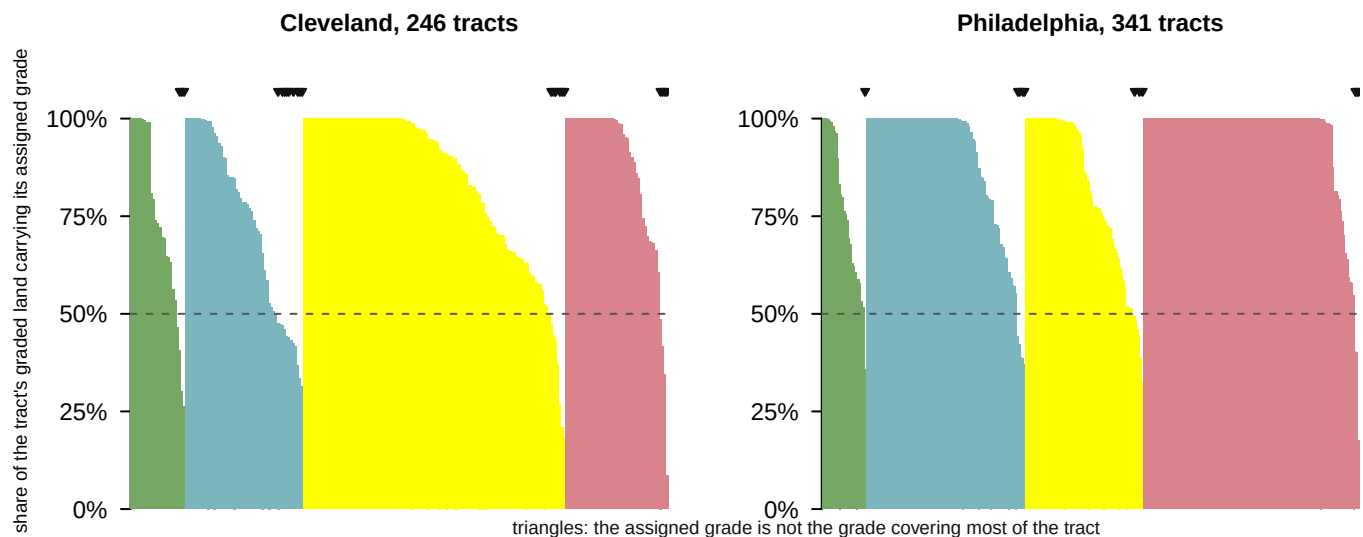


Figure 3. Most tracts are mostly one grade, and a tail of them is not. One bar per tract in Cleveland and Philadelphia, colored by the grade the center point assigned and sorted within each grade. The height is the share of that tract's graded land actually carrying the assigned grade. Marked bars are the 33 tracts of 587 where the assigned grade is not the grade covering most of the tract. The median Cleveland tract stands at 90%.

Most readers, going by Figure 2, write down a fifth or more. It is 33 of 587 – about one tract in 18. The rule is far better than its worst case makes it look, and that is the more uncomfortable finding. A rule that fails loudly can be caught. A rule that fails quietly in 33 places cannot.

The tail is not enormous, but it is not nothing either: 8.9% of Cleveland's tracts are assigned a minority grade, and 3.2% of Philadelphia's. Half of Cleveland's assigned tracts are at least a tenth wrong about their own land.

Three things are worth saying plainly about that, because each of them is a way this chapter could be misread.

Point-in-polygon is the simplest defensible rule, not the only one. A crosswalk is a table that translates one set of areas into another. One weighted by area or by population would assign some of these tracts differently and would not throw the dropped ones away. Whether it would move the within-city results below is not shown here and should not be assumed in either direction.

Figure 3 measures land, not people. A tract split half and half between two grades may have almost everybody living on one side of the old line. The assumption that residents are spread evenly across a tract is doing work in that figure exactly as it does in the numbers it is meant to audit.

The crosswalk is at least reproducible. Re-running the rule from the polygons themselves, rather than reading the grade out of the shipped file, returns the same grade for all but 1 of the 588 tracts in these two cities. That is the least a crosswalk can be asked to be, and it is more than a licensed score behind a login can offer.

07 The answer that arrives first is the wrong one

With the join built, a single row of the resulting file is a strange object: a 2020 census tract carrying a grade assigned to the ground beneath it in the 1930s.

Tract GEOID	State	City	1930s HOLC grade	2020 population	Black residents	% Black
06001400100	CA	Oakland	B	3038	144	4.74
06001400200	CA	Oakland	B	2001	40	2.00
06001400300	CA	Oakland	C	5504	561	10.19

Pool all 4,489 of those rows by grade and the comparison the question seems to call for appears immediately.

HOLC grade	Tracts	2020 population	Black residents	% Black
A	273	978,270	163,372	16.7
B	899	3,269,467	762,053	23.3
C	2096	7,273,922	1,555,044	21.4
D	1221	3,984,323	1,018,080	25.6

Areas graded A are 16.7% Black today; areas graded D are 25.6%. The gap is 8.9 points.

It is a real difference, and after the appraisal sheet at the top of this chapter it is a deflating one. Worse, the ordering does not even behave: areas graded C are 21.4% Black, which is *less* than the 23.3% in areas graded B, so the national gradient is not monotone. If the maps had the reach the historical record suggests, this table should look more decisive than it does.

The table is not measuring what it appears to measure. **It compares A areas in one city against D areas in another.** Cleveland's D areas and Oakland's A areas sit in it together,

and American cities differ enormously in overall Black population for reasons that have nothing to do with the HOLC. San Francisco's D neighborhoods sit in a city that is about 5% Black; Cleveland's A neighborhoods sit in a city that is roughly half Black. Pooling them lets differences *between* cities swamp the difference *within* them, which is the only difference the question is about.

Comparison	What varies	Is it the right test?
Pooled nationally	grade AND city AND region	No
Within each city	grade only	Yes

08 Holding the city fixed

The repair is to make the comparison inside each city, where the same housing market, the same municipal government, the same school district boundaries and the same regional history apply to both grades.

City	A tracts	D tracts	A areas % Black	D areas % Black	Gap
Cleveland	25	47	21.3	68.1	46.9
Pontiac	1	4	32.1	71.0	39.0
Decatur	1	4	19.4	54.8	35.4
Toledo	3	4	6.6	40.5	33.9
Greater Kansas City	5	34	4.0	37.7	33.6
St. Louis	15	17	14.0	47.6	33.6
Chicago	18	249	3.2	36.6	33.4
Pittsburgh	7	43	4.2	34.1	29.9

In Cleveland – the city whose 1939 sheet opens this chapter – neighborhoods the HOLC graded A are 21.3% Black today and neighborhoods it graded D are 68.1%. That is 46.9 points, in one city, between two sets of neighborhoods separated by nothing except a letter a federal appraiser assigned before the Second World War.

Quantity	Value
Cities with both A and D areas	33
Average within-city gap (points)	14.5
Largest gap	46.9 (Cleveland)
Cities where D areas are more Black than A areas	26

Cleveland is the extreme case, and the extreme case is never the argument. Across all 33 cities that have both A and D areas in this file, the average within-city gap is 14.5 points, and 26 of the 33 run in the direction the history predicts. Figure 4 puts every one of them on the same axes.

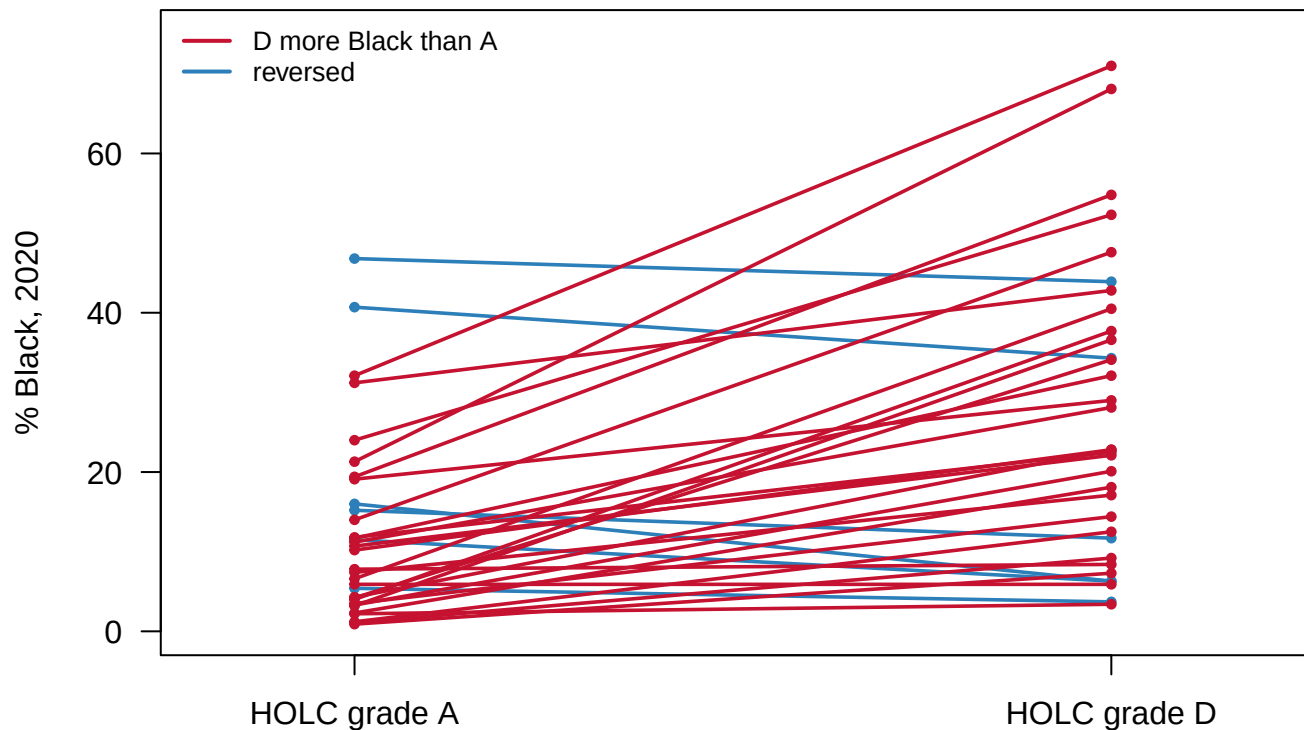


Figure 4. Within each city, from its A-graded ground to its D-graded ground. One line per city, joining the share of residents who are Black in the tracts graded A to the share in the tracts graded D, 2020 Census. Of the 33 cities with both grades, 26 rise and 6 fall; the steepest rise is Cleveland, at 46.9 points. The average line rises 14.5 points.

09 The cities that run the other way

The 6 downward lines in Figure 4 are the most useful thing in this chapter, and the temptation to call them noise should be resisted.

City	A tracts	D tracts	A population	D population	A % Black	D % Black	Gap
Joliet	2	1	6,265	3,040	16.0	6.3	-9.6
Philadelphia	28	138	97,336	548,467	40.7	34.3	-6.4
Stockton	1	6	5,750	26,692	11.5	6.3	-5.2
Canton	1	1	3,278	2,884	15.2	11.7	-3.5
Detroit	22	102	82,787	251,281	46.8	43.9	-3.0
San Jose	1	6	3,728	29,341	5.4	3.7	-1.7

They are two different phenomena wearing the same sign, and separating them is the difference between a finding and an artifact.

Philadelphia is a real reversal. Its D areas are 34.3% Black and its A areas 40.7% – backwards by 6.4 points, across 138 D tracts holding 548,467 people, so it cannot be waved away as thin data. What it reflects is North Philadelphia, graded D and now substantially gentrified, alongside historically Black neighborhoods that were never graded D at all.

The 1937 grade did not determine 2020. A great deal happened in between: urban renewal, deindustrialization, highway construction, and, recently and very unevenly, reinvestment.

Joliet is not a real reversal. Its 9.6-point gap rests on 2 A tracts against 1 D tract. A percentage computed on a handful of tracts is not a stable quantity. Printing it beside Cleveland's 25-against-47, as though the two carry the same weight, would be a way of misleading the reader with entirely accurate numbers. Several of the other reversals are small in the same way.

Taken together, the reversals settle which of three sentences this evidence will bear.

Reading	Does this data support it?
The HOLC grade determines a neighborhood's race today	No — cities reverse
The HOLC grade is associated with it, on average, within cities	Yes — within-city comparison
The HOLC grade caused it	Not by itself

The middle sentence is what 33 cities and 4,489 tracts support. The first is refuted by Philadelphia. The third needs an argument that no cross-section of this kind can supply, and it happens to be the argument the field has spent the most effort on.

10 Did the maps do anything, or only record what was already there?

The skeptical position is serious and should be stated at its strongest. The appraisers were describing segregation that already existed, much of it produced by restrictive covenants, by violence, and by private lending practices that predated the HOLC by decades. On that reading the maps are an excellent *record* of 1930s segregation and no evidence at all about its causes. The within-city gap in Figure 4 would show only that what was already there stayed there.

Aaronson, Hartley and Mazumder (2021) addressed exactly this. The skeptic objects to comparing graded neighborhoods with ungraded ones, so they did something else. Only cities above a population threshold were mapped at all, so they compared similar neighborhoods in cities that just barely were mapped and just barely were not. They found effects on segregation, on homeownership, on house values and on credit access that persisted for decades, and concluded that the maps had causal consequences rather than merely descriptive ones.

Rothstein's *The Color of Law* (2017) makes the wider historical case that American residential segregation was largely the product of explicit government policy rather than private preference. The HOLC was one tool among several, alongside FHA underwriting standards and the siting of public housing.

What remains genuinely disputed is not whether the maps mattered, but how much and by what route. How much of today's pattern comes from the HOLC itself, and how much from the long series of later policies that followed the same lines because the lines were already drawn.

11 What the ground can testify to

The evidence assembled here is unusually direct in one respect and unusually limited in another, and both deserve stating without hedging.

It is direct because nothing in it is inferred from a model. The historical source is the federal document itself, digitized with its appraisal text intact. So the racial reasoning in the opening pages is quoted rather than reconstructed. The demographic source is a full enumeration rather than a sample. And the link between them is visible. It is a rule stated in one sentence, with its failures counted in Figure 3, and its output rebuildable from the polygons to within 1 tract of the file it was checked against.

It is limited because a great deal rests on that one sentence. The center-point rule gives 33 of 587 tracts in Cleveland and Philadelphia a grade that is not the grade covering most of their land. It also drops 181 tracts that overlap a graded area but whose center point does not. The file covers 56 cities in 6 states, which is not the country. It is a single snapshot, so it can show a difference in 2020 but not a path. These neighborhoods may have been converging for fifty years, or diverging, or crossing.

Cities with one or two A tracts sit in the same tables as cities with dozens. And grade was never the only thing that differed between an A neighborhood and a D neighborhood in 1937. That is precisely why the causal question above needed a different research design, rather than a bigger version of this one.

So, within a city: neighborhoods the federal government graded “Hazardous” in the 1930s are on average 14.5 points more Black today than neighborhoods the same government graded “Best”. That pattern is still legible after eighty years and several complete rebuildings of American cities. The pooled national comparison that suggests otherwise is measuring cities, not grades. Neither of those statements establishes that the grades caused the pattern, and the 6 cities that run the other way are a standing reminder that the map did not fix anything in place permanently.

Three questions are left genuinely open, and none of them is going to be settled by more of this table. How much of today’s segregation is HOLC-attributable rather than produced by the later policies that traced the same lines. Whether reinvestment reverses the pattern, as Philadelphia suggests it can, and for whom, and at what cost to the people already living there. And what any of it implies for remedy: documenting a durable effect of an explicit federal policy does not, by itself, identify what would undo it.

The appraiser in Garfield Heights ended his paragraph with a prediction about who would and would not be living on those streets. What can be said now is not that he was right, and not that he made himself right. It is that the boundary he drew is still visible in the census – which is a strange fate for a piece of municipal paperwork, and the reason it is worth reading the paperwork.

12 Sources

Data

- Nelson, R. K., Winling, L., Marciano, R., Connolly, N. D. B., et al. *Mapping Inequality: Redlining in New Deal America*. Digital Scholarship Lab, University of Richmond, CC BY-NC-SA. <https://dsl.richmond.edu/panorama/redlining/> – **the HOLC polygons, the grade colors and the appraisal text drawn and quoted above are theirs**. Retrieved

2026-08-10: the national area file (`static/mappinginequality.json`, 10,563,682 bytes, 10,154 areas across 314 city surveys, of which 9,324 carry a grade A to D) and the area description sheet for OHCleveland1939 area D9. The maps and forms underneath are 1930s US federal government records.

- U.S. Census Bureau, 2020 Decennial Census, the P.L. 94-171 redistricting file, one state at a time. https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/
- U.S. Census Bureau, TIGER/Line 2020 census tract boundaries — the outlines drawn over the 1939 areas in Figure 2.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`cities.csv`, `fig_holc_attr.csv`, `fig_holc_meta.csv`, `fig_holc_rings.csv`,
`fig_mix.csv`, `fig_quote.csv`, `fig_source.csv`, `fig_zoom_attr.csv`, `fig_zoom_holc.csv`,
`fig_zoom_tracts.csv`, `fig_zoom_win.csv`, `tracts.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `cities.csv` gives a gap for each city: the share Black in A-graded areas against the share in D-graded areas. Sort every city by it. Which city has the smallest gap, and is that good news?
- `tracts.csv` attaches a 1930s grade to present-day tracts. Group by grade and compare `pct_black`. Then ask what a tract drawn in 2020 has to do with a line drawn in 1939.
- `fig_mix.csv` has `dom_share`, how much of a tract sits inside a single grade. Find the tracts where it is low. Those are the ones any map of this data draws wrong.
- `fig_quote.csv` holds one appraiser's own words about one neighborhood. Read them. Then look up what that neighborhood is like now.

Academic literature

- Aaronson, D., Hartley, D. and Mazumder, B. (2021). "The Effects of the 1930s HOLC 'Redlining' Maps." *American Economic Journal: Economic Policy* 13(4): 355-392.
- Rothstein, R. (2017). *The Color of Law: A Forgotten History of How Our Government Segregated America*. Liveright.

How the figures were made

The HOLC polygons were fetched from the address above and measured against the 2020 tract boundaries. Every URL is recorded, with the date it was retrieved and the feature counts that came back, and it is the only place here that touches a shapefile. Every number printed above is computed from the files it wrote, at the moment this document was rendered.

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original sources, without any starting files. Please do the following and show your work.

THE FIRST SOURCE. "Mapping Inequality: Redlining in New Deal America", Digital Scholarship Lab, University of Richmond (CC BY-NC-SA). One GeoJSON file, about 10.5 MB, holding every area drawn on the Home Owners' Loan Corporation's 1930s security maps:
<https://dsl.richmond.edu/panorama/redlining/static/mappinginequality.json>
No account or key is needed. Each feature is one appraisal area with its city, state, and grade – 10,154 areas, of which 9,324 carry a grade A through D. The grade field arrives padded with spaces: the same grade appears as "A", "A ", and " A". Trim it before filtering, because keeping only A through D without trimming looks like a clean subset and silently drops areas.

THE SECOND SOURCE. U.S. Census Bureau, TIGER/Line 2020 census-tract boundaries, one zip per state, at
https://www2.census.gov/geo/tiger/TIGER2020/TRACT/tl_2020_<FIPS>_tract.zip
for six states: California (06), Illinois (17), Michigan (26), Missouri (29), Ohio (39), Pennsylvania (42). Every tract polygon carries an internal point the Bureau guarantees to lie inside the tract.

THE THIRD SOURCE. 2020 Census P.L. 94-171 redistricting data for the same six states – table P1 has each tract's total, white, and Black population. One zip per state, named like Michigan's:
https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File-PL_94-171/Michigan/mi2020.pl.zip

The files inside are pipe-delimited segments that join on a record number. Read that number as text on both sides – it is zero-padded, and a numeric read makes the join silently match nothing.

WHAT TO BUILD.

1. A grade for each 2020 tract: the tract takes the grade of the graded area its internal point falls inside, first match wins. Keep the tracts that get one.
2. Those graded tracts joined to their 2020 population counts, with percent Black per tract.
3. For every city holding both an A tract and a D tract, the percent Black of the pooled population on each side, and the gap between the two.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- 10,154 areas in the map file, 9,324 of them graded A through D
- 4,489 tracts carrying a grade across the six states
- 33 cities with both an A and a D tract; the widest gap is Cleveland, whose once-A tracts hold 79,609 people and whose once-D tracts hold 100,969

Mapping Inequality keeps adding newly digitized cities; a larger area count means the project grew, not that you made an error.